

JEFFERY LIU

Github: liujeffery — LinkedIn: liujeffery — (647)-548-8664 — jefferyliu521@gmail.com

EDUCATION

McMaster University

September 2021 - April 2027

B.E. in Software Engineering, 3.85 GPA

Hamilton, ON

Relevant courses: Data Structures & Algorithms, OS & Concurrency, Databases, Digital Systems & Interfacing, OOP

SKILLS

Languages: Python, Java, Javascript, Typescript, C, C++, HTML, CSS, Matlab, SQL

Technologies: Tensorflow, Large Language Models (LLMs), Keras, Computer Vision, PyTorch, REST API, Docker, Version Control (Git), Object-Oriented Programming (OOP), Data Structures & Algorithms, Github Actions (CI/CD), React, Node.js, Flask, Unit Testing, Natural Language Processing (NLP), Agile/Scrum, Bash, Linux, Electron

EXPERIENCE

Hydro One

May 2025 - August 2026

Software Intern — *Python, SQL, HuggingFace, LLMs, NLP, Selenium, Visual Basic, Git*

Toronto, ON

- Built a tool using **Python** and **quantized LLMs** to preprocess, batch, and generate summaries from shift logs, reducing manual work by **90 hours a month** and **token-related costs by 60%**
- Used **SQL** and **Python** to automatically compile **15+ monthly reports** and eliminate a manual ticketing system, reducing work by **30 hours a month**
- Created a **web-based script** using **Selenium** to automate bulk changes to **1000+ devices**, reducing manual work by **40 hours a month** and eliminating **\$1000 a month** in third-party fees
- Consolidated **over 100 pages** of scattered documentation into a single, hyperlinked wiki, slashing file volume by **60%** and reducing information retrieval time by **50%**

Healthcare Systems R&A

May 2023 - April 2024

Software Engineering Intern — *Python, Tensorflow, Keras, PyTorch, DGL, Git*

Toronto, ON

- Created core functionality features for critical company projects worth over **\$700,000** by developing **convolution, recurrent, and LSTM machine learning models** using **Tensorflow, Keras, and PyTorch**
- Reduced development time by **40%** by spearheading implementation of **end-to-end pipelines** for **core landmark detection models** using **Python**
- Engineered data preprocessing pipeline (**imputation, feature extraction**) in NumPy to process over **1,000,000** data signals, cutting processing time by **90 hours/month**
- Developed a **graph neural network** for optimized feature extraction using DGL, improving end model accuracy by **25%** and reducing computational demand of model by **30%**

Canadian Imperial Bank of Commerce

May 2022 - August 2022

Technical Analyst — *Python, Java, ServiceNow, Atlassian Jira, Agile/Scrum, Git*

Toronto, ON

- Reduced sector-wide permissions search times by **75%** by creating a **custom internal management tool** and implementing asynchronous updates, using **Java** and **Atlassian Jira** workflow
- Eliminated **45 hours a month** of team-wide tasks on critical server by replacing the existing manual check with an automated service notification project using Python

TECHNICAL PROJECTS

McMaster Interdisciplinary Satellite Team

May 2023 - present

Software Specialist — *Javascript, Python, C/C++, Typescript, Github Actions, Docker, Git, Flask*

Hamilton, ON

- Developed control centre software in **React.js** and **Flask** to receive, visualize, and send data in real-time
- Consolidated disjointed legacy APIs into a lightweight **JSON RPC 2.0** bridge between **Python** and **Electron**, reducing **maintenance and development times by 66%** and computational overhead by **50%**
- Standardized firmware unit testing between **Linux, Windows, and macOS** environments by containerizing the test environment in **Docker** and deploying it via **Github self-hosted runners**, reducing test setup time by **70%**
- Built a **CI/CD pipeline** with **Github Actions** to automate C/C++ **unit testing**, reducing **40 hours** of work monthly